

BioFlux™ 1000Z

Automated imaging for live cell analysis under shear flow

A complete solution for high-content screening under shear flow:

Higher biological relevance:

BioFlux 1000Z provides controllable shear flow for high-content imaging experiments. Bridges the gap between in vitro and in vivo screening and research.

Fully-integrated microscopy:

BioFlux 1000Z comes complete with all the components necessary for high resolution microscopy imaging in brightfield, phase, fluorescence, and DIC. Built around the industry-leading ZEISS Axio Observer microscope platform. Offers powerful features for imaging, including autofocus, automated stage, and shear flow control.

Convenience of single point control: All hardware and analysis controls run through the BioFlux Montage Software. Makes it simple and easy to coordinate all of your screening experiments and analysis from a single user interface.

Increased throughput: Runs up to 96 shear flow assays in parallel, making it simple to screen hundreds of data points per day. Automated stage and image acquisition provide rapid scanning of experimental channels.

Unattended operation: Simple-to-use experiment scheduler allows unattended imaging and overnight assays for kinetic and time-lapse experiments.

The BioFlux 1000Z provides a high-content imaging platform for running physiologically-relevant shear flow assays. The fully-integrated and automated system delivers high-resolution microscopy data with the ultimate in throughput, convenience and flexibility.

The BioFlux 1000Z System incorporates Fluxion's proprietary Well Plate Microfluidic™ technology with an automated microscopy system to enable high throughput shear flow assays. It offers the biological relevance of a laminar flow cell with the throughput and convenience of standard microplates. The fully-integrated system comes complete with all the tools necessary to generate your data quickly and reliably.

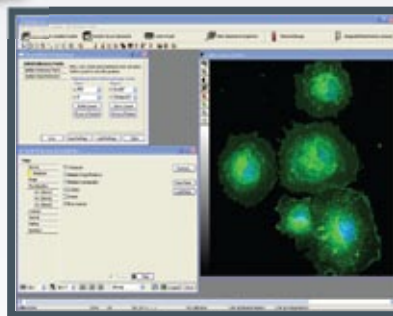
BioFlux 1000Z System Overview:

BioFlux 1000Z is a fully-integrated high content screening platform for bridging the gap between in vitro and in vivo experiments. It integrates a high performance microscopy workstation with the BioFlux electropneumatic pumping system for controlling shear flow. BioFlux Montage software offers single point control for designing experiments, acquiring images, and analyzing data.



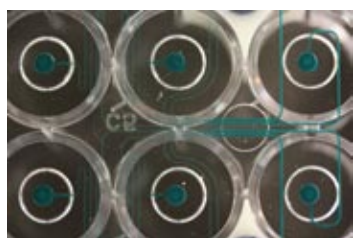
BioFlux Controller delivers programmable shear flow to BioFlux plates. Covers a wide shear range (0.1–200 dyne/cm²) to address many physiological applications.

BioFlux Montage Software delivers single point control for all system components. Design experiments, acquire images, and analyze data all from one intuitive software package.



Customized automated stage enables fast scanning of BioFlux Plates with industry-leading speed, precision and reliability. Stage positions for BioFlux plates come pre-configured for plug and play operation.

Automated microscopy station provides high performance imaging in brightfield, phase, DIC, and fluorescence. Cooled sCMOS camera provides high resolution, low noise data acquisition.



Product Specifications

BioFlux Controller:

Shear flow range: 0.1–200 dyne/cm²
 Temperature Control: ambient to 50°C
 Dimensions: 12" (W) X 13" (L) X 9" (H)
 30cm (W) X 33cm (L) X 22cm (H)

Inverted Microscope:

Microscope: ZEISS Axio Observer 7
 Illumination: High intensity LED
 Objectives: 5X–100X
 Focusing: automated Z-drive focus,
 software-based autofocus

sCMOS Camera (standard):

Imaging array: 2048x2048, 6.5µm
 pixels, 16-bit digitization
 Quantum efficiency: 82%
 Speed: 30 frames/s

Automated Stage:

XY Travel: 130 x 100mm Repeatability:
 <1µm
 Resolution: 0.05µm
 Sample holder: customized interface
 for BioFlux plates

Automated Fluorescence:

Lamp: 120W metal halide
 Filters: 6-position motorized filter
 turret Automated integrated shutter

BioFlux Montage Software:

Drivers: BioFlux controller, camera, stage,
 focus, filter turret changers
 Modules: BioFlux control, multi-dimensional
 acquisition, autofocus, analysis

Desktop Computer:

Processor: Intel Core i7
 Operating system: Windows 10
 Memory: 32GB, 1TB HD, 256GB
 SSD



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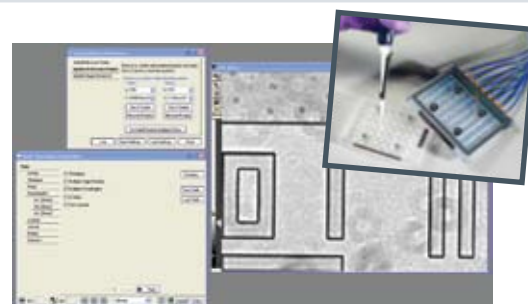
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An automated workflow designed for throughput and convenience

BioFlux 1000Z offers a simplified approach to running shear flow assays. The intuitive and easy to use software package provides complete control over shear flow, image acquisition and data analysis.

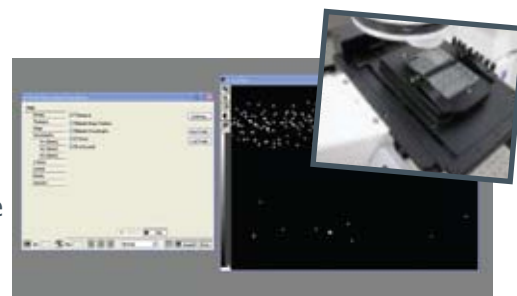
Setup

Load cells/reagents in to BioFlux plate
 Set calibration points
 Select acquisition parameters



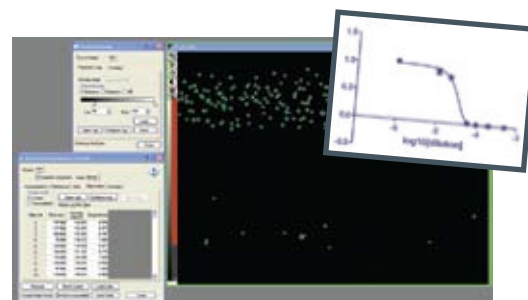
Run

Select flow protocol
 Acquire images using:
 Time lapse or endpoint
 Brightfield, phase, DIC, or fluorescence



Analyze

Autoselect and count cells
 Full morphology profile
 Numerous analysis modules available
 Live/dead
 Cell tracking
 Nuclei countingand more



Typical Applications:

- Leukocyte adhesion assays
- Platelet adhesion assays
- Anti-microbial compound screening
- Stem cell assays for kinetics and differentiation
- Pharmacology / dose response assays
- Life cycle analysis (mitosis, apoptosis, etc.)

Ordering Information:

To request a quote or to place an order:
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